1. **Types of Machine Learning**

Supervised Learning: Learns from labeled data (input-output pairs). Goal is prediction.

Unsupervised Learning: Works on unlabeled data, finding hidden patterns or structures.

Reinforcement Learning: Learns by interacting with an environment, receiving rewards or penalties.

2. **Supervised Learning**

Process:

Training: Learn a model from labeled data.

Testing: Evaluate model on unseen data.

Goal: Predict class (categorical) or value (continuous).

Examples: Neural Networks, SVMs, Decision Trees, Bayesian classifiers.

Applications: Predicting disease, loan approvals, spam detection.

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3. **Unsupervised Learning**

Data: Unlabeled, no predefined classes.

Goal: Identify patterns, clusters, or hidden structures. Techniques:

Clustering: Grouping data points (e.g., K-Means, Hierarchical).

Association: Discovering relationships between variables.Applications: Customer segmentation, market basket analysis, anomaly detection.

4. **Reinforcement Learning**

Concept: Agent interacts with environment, takes actions, receives feedback.

Feedback: Reward (positive) or penalty (negative).

Goal: Maximize cumulative reward over time.

Applications: Robotics, game playing (e.g., AlphaGo), self-driving cars.

5. **Classification**

Definition: Assigns data points to one of several categories.

Key Idea: Mutually exclusive C exhaustive categories (each data point belongs to exactly one class).

Process:

Model Construction: Train on labeled data.

Model Usage: Predict classes for new data.

Algorithms: Decision Trees, K-Nearest Neighbors (KNN), Logistic Regression, Naive Bayes.

Applications: Spam filtering, credit approval, medical diagnosis.

6. **Regression**

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Goal: Predict continuous numeric values.

Types:

Linear Regression: Fits a line (y = mx + b) minimizing squared error.

Logistic Regression: Despite the name, used for binary classification (outputs probability).

Applications: House price prediction, stock forecasting, demand prediction.

7. **Evaluation Metrics**

For Classification:

Accuracy: Correct predictions ÷ total predictions.

Precision: Correct positive predictions ÷ total predicted positives.

Recall (Sensitivity): Correct positive predictions ÷ actual positives.

F1-Score: Harmonic mean of precision and recall.

For Regression:

Mean Squared Error (MSE): Average of squared prediction errors.

Root Mean Squared Error (RMSE): Square root of MSE.

Mean Absolute Error (MAE): Average of absolute differences.

R^2 (Coefficient of Determination): Measures variance explained by the model.

Validation Techniques:

Train-Test Split: Dividing data into training (70–80%) and testing (20–30%).K-Fold Cross Validation: Splits data into k parts; trains on k-1 folds, tests on the remaining.

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